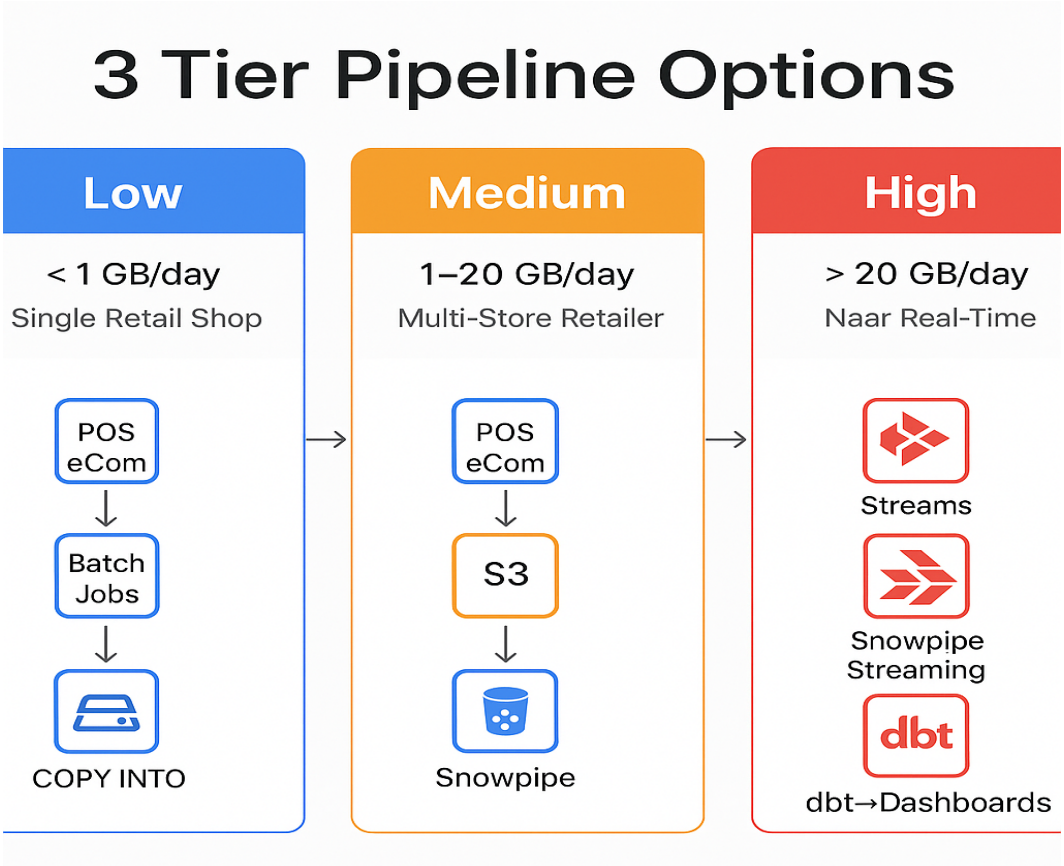
Retail Data Engineering – Technical Blueprint

A complete visual and technical guide for your startup's pipelines, client tiers, architecture, and operational model.

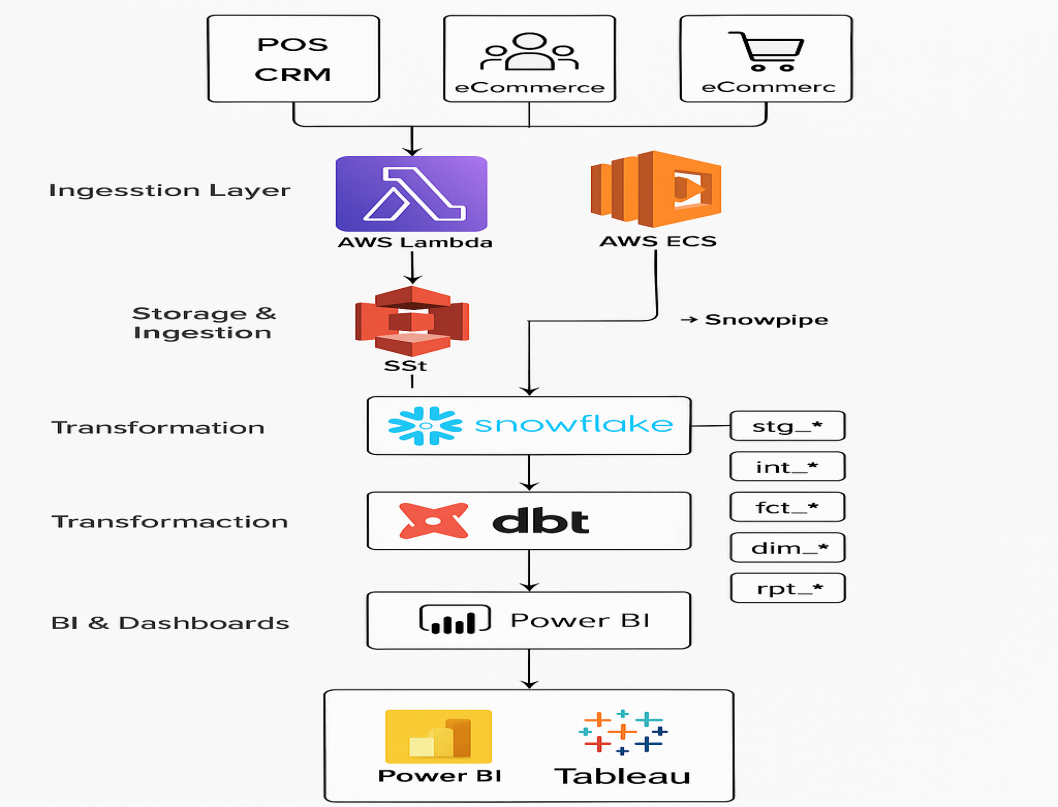
1. Three Tier Pipeline Options

Your platform supports three categories of retail clients based on data volume and latency needs. Each tier uses the simplest pipeline needed—avoiding over-engineering while remaining scalable.



2. Multi-Tenant AWS + Snowflake + dbt Architecture

This architecture supports scalable onboarding of multiple retail clients while ensuring data isolation, cost efficiency, and flexible ingestion patterns (batch, Snowpipe, streaming).


POS
CRM
eCommerce
eCommerc
Ingesstion Layer
AWS Lambda
AWS ECS
Storage &
Ingestion
SSt
→ Snowpipe
Transformation
snowflake
stg_*
int_*
Transformaction
dbt
fct_*
dim_*
rpt_*
BI & Dashboards
Power BI
Power BI
Tableau

3. Detailed Breakdown of Each Tier

Tier 1 – Low Scale
• Data Volume: <1 GB/day
• Latency Needs: Daily/hourly
• Ingestion: Batch → S3 → COPY INTO Snowflake
• dbt: Daily runs
• Best For: Small shops, boutiques, single-store retailers

Tier 2 – Medium Scale
• Data Volume: 1–20 GB/day
• Latency Needs: 15–60 minutes
• Ingestion: Frequent batches + Snowpipe auto-ingest
• dbt: Every 15–30 minutes
• Best For: Multi-store retail, supermarkets

Tier 3 – High Scale
• Data Volume: 20GB+/day or high event throughput
• Latency Needs: 1–5 minutes (near real-time)
• Ingestion: Streaming → S3 → Snowpipe Streaming
• dbt: Incremental via Snowflake Tasks
• Best For: Enterprise chains, high-volume online platforms

4. 30-Day Technical Training Roadmap (Summary)

Week 1: AWS setup, Snowflake setup, dbt config, raw S3 zone, dummy ingestion.
Week 2: Real ingestion jobs, Snowpipe setup, dbt staging models, logging & monitoring.
Week 3: Canonical model design, incremental fact tables, CI/CD for dbt, documentation.
Week 4: Dashboard templates, alerts, lifecycle rules, final QA, first client demo readiness.

Your team becomes production-ready by the end of Week 4.

5. Tools & Skills Required (Onshore + Offshore)

Onshore (You):
AWS architecture, Snowflake design, dbt modelling patterns, canonical schemas, KPI definition, client communication.

Offshore Team:
Python (APIs), Lambda/ECS, S3 ingestion, Snowpipe configuration, dbt development, dashboard building, monitoring & alerts.

Shared Tools:
GitHub, dbt Core, Power BI/Tableau, CloudWatch, Jira/Trello, Slack/Teams.

6. Final Summary

You now have a fully documented architecture strategy, tiered client offering, 30-day training roadmap, and visuals. This enables you to approach retailers confidently, build repeatable pipelines, onboard offshore team members, and scale to 10–30 clients without rewriting your architecture.